College Attendance Analytics Dashboard

Power BI Project Report

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Project Title	College Attendance Analytics Dashboard
Tool Used	Microsoft Power BI Desktop (Power Query, DAX, Report View)

1. Dataset Description

The project analyses college attendance for **245 students** across **four academic years** (70, 50, 60 and 65 students in the 1st to 4th year respectively) and three departments: **CSE, AI&DS and IoT**. Attendance is tracked for **10 subjects** (6 theory and 4 practical/lab) for the period **6 July 2026 to 13 November 2026**. The data model uses a session-level attendance table, *FactAttendance*, two dimension tables, *DimStudent* and *DimSubject*, and a student-level summary table, *AttendanceSummary*.

Field / Item	Description
FactAttendance	Main attendance table; one record per student attendance entry for a class session
DimStudent	Student dimension table with student details
DimSubject	Subject dimension table with subject details
AttendanceSummary	Student-level attendance summary table
StudentID	Student identifier, used to count distinct students (*Total Students*) and to evaluate attendance per student
RegistrationID	Student registration number (e.g. in the form PIET23IT016, as displayed in the dashboard tables)
StudentName	Name of the student
Department	Student department: CSE, AI&DS or IoT
AcademicYear	Academic year of the student (1st to 4th year)
Section	Class section of the student
Gender	Student gender (122 male and 123 female students shown on the Students page)
SubjectCode	Subject identifier (e.g. AOA, CD, DMTC, DVR, OS, PDS, with -LAB codes for practical subjects)
SessionDate	Date of the class session
SessionNumber	Session number; together with SessionDate it identifies a unique class (see *Total Classes*)
PresentFlag	Numeric attendance indicator; summed to count present records in *Overall Attendance %*

Other attributes visible in the dashboard visuals include subject name, subject type (Theory/Practical), student attendance percentage, attendance status (Safe / Warning / Critical), shortage rank, and classes required. The tables in which these are stored have not been verified and are therefore not listed above.

2. Objectives

- Build an interactive college attendance dashboard using Microsoft Power BI.
- Monitor overall attendance and attendance trends over the period July to November 2026.
- Compare attendance by department, academic year, student and subject.
- Identify students below 75% attendance and critical students below 65% attendance.
- Identify subjects with attendance shortages.
- Estimate the additional classes required to reach the 75% attendance target.
- Enable interactive exploration using slicers and dashboard page navigation.

3. Data Cleaning Performed

Data preparation was carried out in Power Query before the visuals were built. The table below separates what is evident from the project material from the validation checks that are recommended for the dataset. Recommended checks are **not** claimed as completed operations.

3.1 Evident from the project material

- Attendance data is organised into the tables *FactAttendance*, *DimStudent*, *DimSubject* and *AttendanceSummary*.
- *PresentFlag* is used numerically (summed) in the attendance measure, so it must be stored as a numeric column.
- *SessionDate* and *SessionNumber* are used together to identify unique class sessions.
- Attendance categories use consistent thresholds across pages: Safe (75% or above), Warning (65% to 74.9%) and Critical (below 65%).
- Months on the monthly trend chart appear in calendar order (July to November).

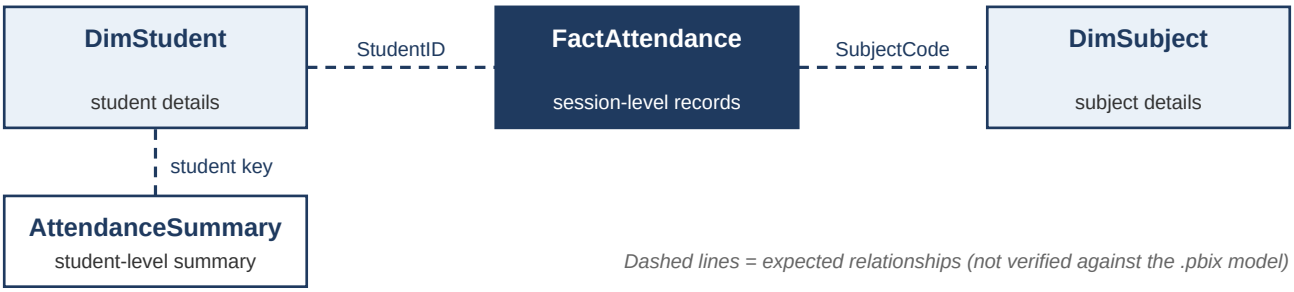
3.2 Recommended validation checks

- Check column names and data types in Power Query for all four tables.
- Ensure *SessionDate* uses a date data type.
- Keep *StudentID* and *RegistrationID* in a suitable data type (identifiers should not be treated as quantities).
- Check for blank values and duplicate attendance records (for example, repeated student, subject, date and session combinations).
- Check that every student and subject code in *FactAttendance* has a matching record in *DimStudent* and *DimSubject*.
- Maintain the same attendance category thresholds in every measure and visual.

4. Data Model

The report is built on a star-style model with one session-level fact table and two dimension tables, plus a student-level summary table.

Table	Role in the model
DimStudent	Student details and student-level categories (department, academic year, section, gender).
DimSubject	Subject details and subject classification (theory or practical).
FactAttendance	Session-level attendance records (student, subject, date, session number, present flag).
AttendanceSummary	Student-level attendance summaries, where applicable.



Dashed lines = expected relationships (not verified against the .pbix model)

Expected relationships: *DimStudent* to *FactAttendance* through the student identifier, and *DimSubject* to *FactAttendance* through *SubjectCode*. These are expected relationships based on the field names and the DAX measures; the relationship cardinality and filter direction have not been verified in the Power BI model view.

5. DAX Measures

The KPI cards and visuals are driven by the following DAX measures, reproduced exactly as supplied for this project.

5.1 Overall Attendance %

```
Overall Attendance % =
DIVIDE(
    SUM(FactAttendance[PresentFlag]),
    COUNTROWS(FactAttendance),
    0
)
```

Calculates the proportion of present attendance records among all attendance records in the current filter context. The result is a fraction (for example 0.813) displayed as a percentage. The denominator is the number of **attendance records**, not the number of unique classes.

5.2 Total Students

```
Total Students =
DISTINCTCOUNT(DimStudent[StudentID])
```

Counts the distinct students in the current filter context (245 with no filters applied).

5.3 Students Below 75%

```
Students Below 75% =
COUNTROWS(
    FILTER(
        VALUES(DimStudent[StudentID]),
        CALCULATE([Overall Attendance %]) < 0.75
    )
)
```

Evaluates the attendance percentage of each student in the current filter context and counts the students whose attendance is below 75%.

5.4 Critical Students

```
Critical Students =
COUNTROWS(
    FILTER(
        VALUES(DimStudent[StudentID]),
        CALCULATE([Overall Attendance %]) < 0.65
    )
)
```

Counts the students whose attendance is below the 65% critical threshold.

5.5 Total Classes (unique date/session)

```
Total Classes =
COUNTROWS(
    SUMMARIZE(
        FactAttendance,
        FactAttendance[SessionDate],
        FactAttendance[SessionNumber]
    )
)
```

Counts the distinct combinations of *SessionDate* and *SessionNumber*, i.e. the number of unique class sessions.

5.6 Percentage of Students at Risk

```
Percentage of Students at Risk =
DIVIDE([Students Below 75%], [Total Students], 0)
```

Expresses the students below 75% as a share of all students. With the dashboard figures (58 of 245) this gives approximately 23.7%.

> **Note on counting units.** *Overall Attendance %* works on **attendance records** (one record per student per session), whereas *Total Classes* counts **unique date/session combinations**. These are different units and must not be interpreted as the same quantity. Also, the Attendance page card labelled “Total Classes” displays about 65K, which equals its present (about 53K) plus absent (about 12K) counts, i.e. a record-level total. The measure that drives that card is not among those supplied, so the card should be read as a record count.

6. Dashboard Screenshots

The report contains five pages, navigated through the buttons on the left of each page: Overall, Students, Subjects, Attendance and Shortage.

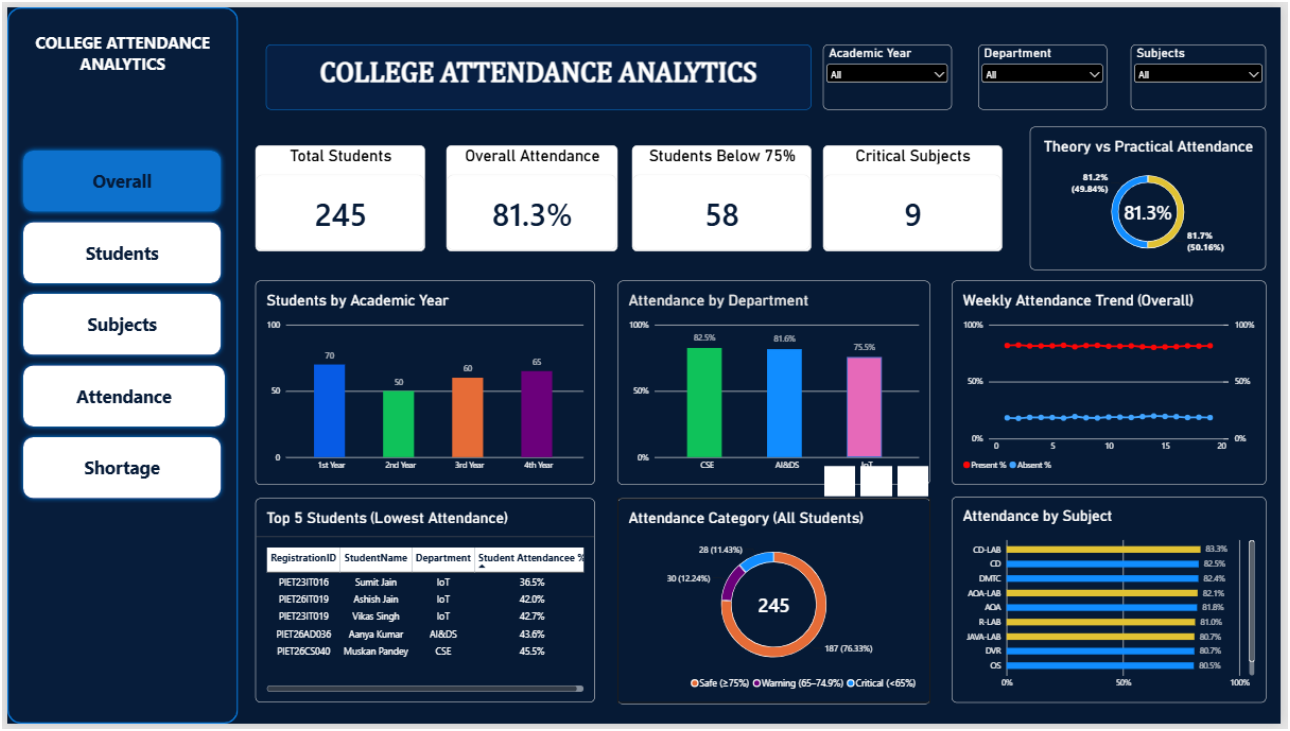


Figure 1: Overall Attendance Analytics Dashboard

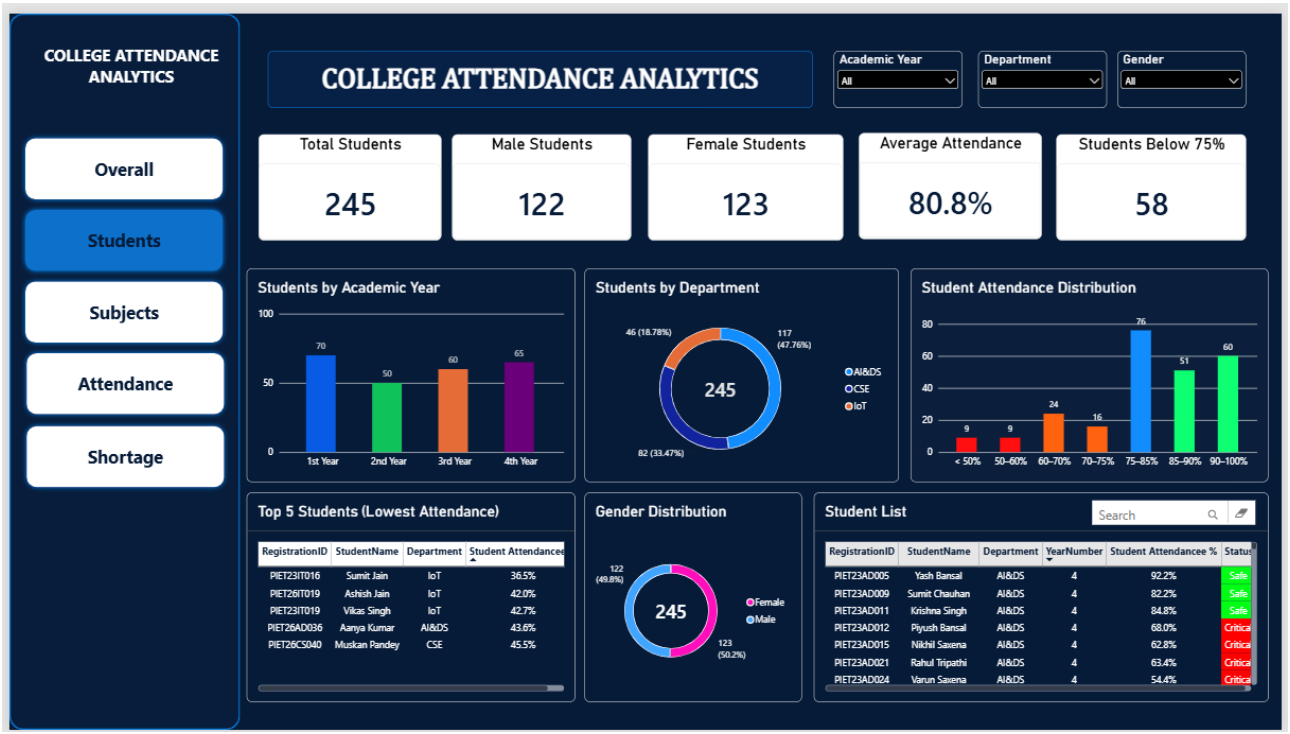


Figure 2: Student Analytics Dashboard

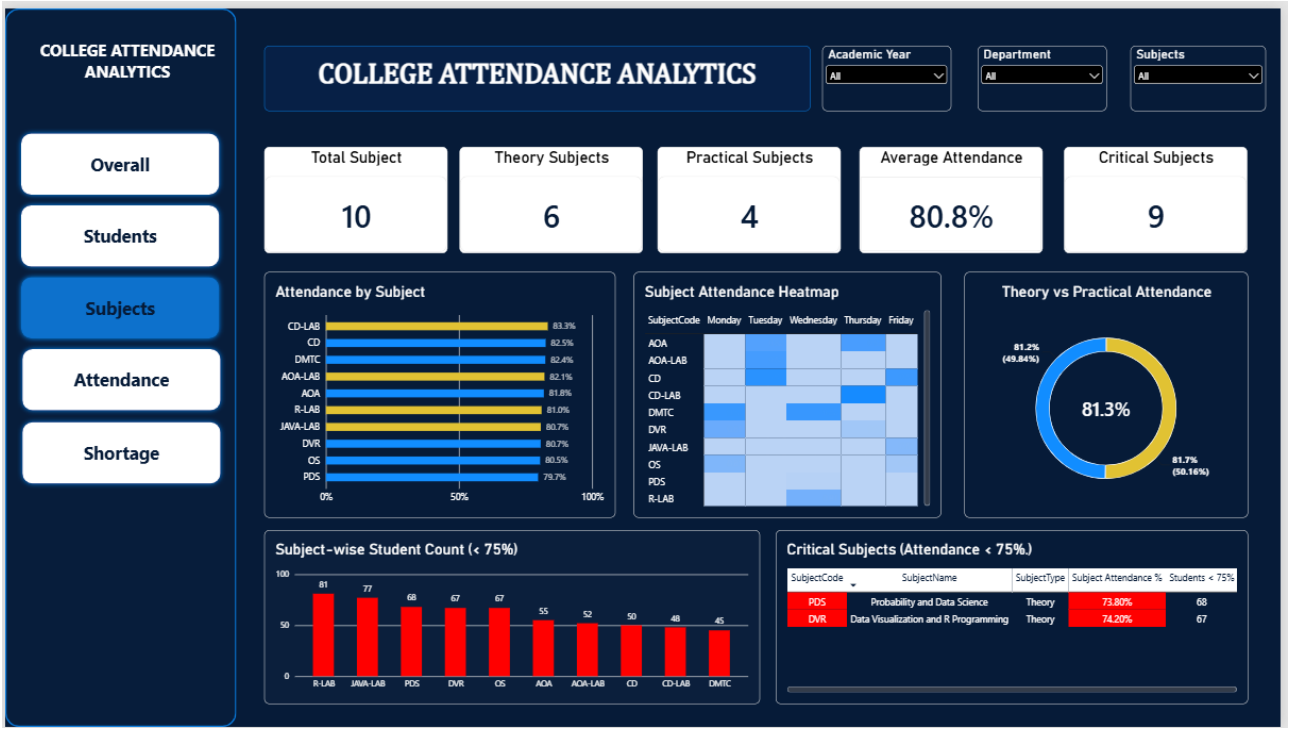


Figure 3: Subject-wise Attendance Dashboard

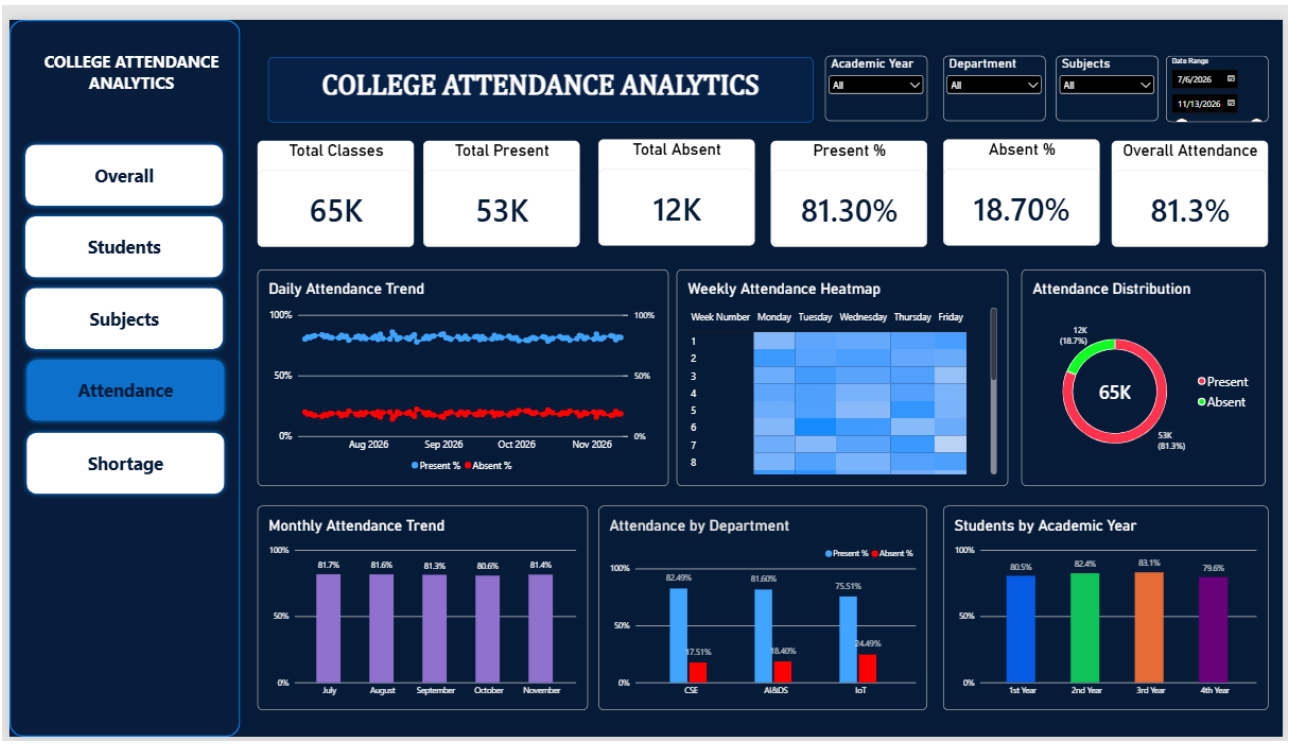


Figure 4: Attendance Trends Dashboard

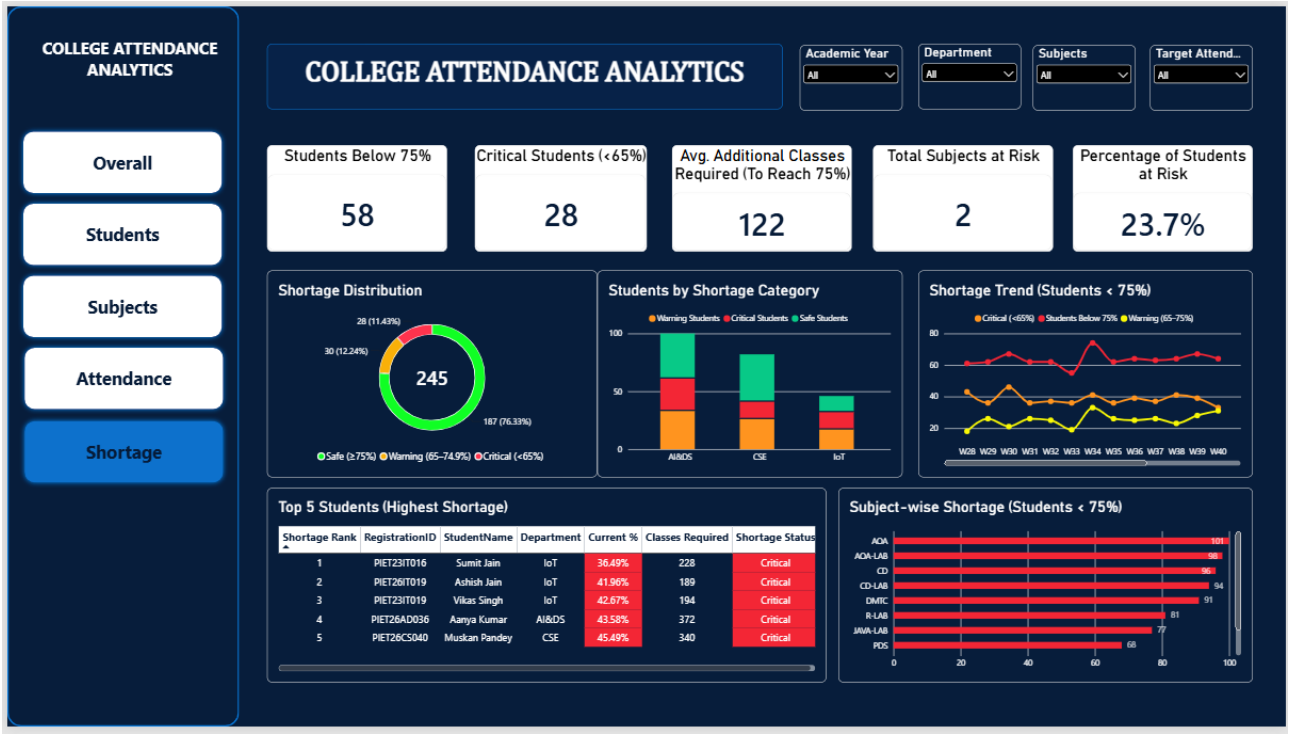


Figure 5: Attendance Shortage Analysis Dashboard

7. Key Findings

All values below are reported as displayed on the dashboard pages.

Indicator	Value shown	Page
Total students	245	Overall / Students
Overall attendance	81.3%	Overall / Attendance
Students below 75%	58 (23.7% at risk)	Overall / Shortage
Critical students (below 65%)	28	Shortage
Attendance categories	Safe 187 (76.33%), Warning 30 (12.24%), Critical 28 (11.43%)	Overall / Shortage
Attendance records	about 65K (about 53K present, about 12K absent)	Attendance
Subjects	10 (6 theory, 4 practical/lab)	Subjects
Avg. additional classes required (to reach 75%)	122	Shortage

- **Overall:** the dashboard shows 245 students, 81.3% overall attendance, 58 students below 75% and 9 critical subjects.
- **Students:** 122 male and 123 female students. Department headcounts are AI&DS 117, CSE 82 and IoT 46. The distribution chart shows 58 students in the bands below 75% (9 below 50%, 9 at 50–60%, 24 at 60–70% and 16 at 70–75%).
- **Departments:** displayed attendance is CSE 82.5%, AI&DS 81.6% and IoT 75.5%. IoT is the lowest of the three.
- **Academic years:** displayed attendance is 80.5% (1st year), 82.4% (2nd), 83.1% (3rd) and 79.6% (4th).
- **Trend:** monthly attendance is shown as 81.7% (July), 81.6% (August), 81.3% (September), 80.6% (October) and 81.4% (November); the daily and weekly trend lines appear relatively flat.
- **Subjects:** the subject attendance chart ranges from 83.3% (CD-LAB) to 79.7% (PDS). The Critical Subjects table highlights PDS (73.80%, 68 students below 75%) and DVR (74.20%, 67 students).

- **Lowest attendance:** the five students with the lowest attendance range from 36.5% to 45.5%; the Shortage page lists the classes each would need to reach 75% (from 189 to 372).
- **Shortage:** the Shortage page reports 58 students below 75%, 28 critical students, 122 average additional classes required, 2 subjects at risk and 23.7% of students at risk, with subject-wise shortage counts for each subject.

Notes on values that differ between pages. Some figures are not identical across pages and are reported as displayed; they may reflect different measures, filters or calculation contexts rather than errors. For example: (a) overall attendance is 81.3% while the Students and Subjects pages show an average attendance of 80.8%; (b) the Subjects page shows PDS at 79.7% in the subject chart but 73.80% in the Critical Subjects table, and DVR at 80.7% versus 74.20%; (c) the Overall and Subjects pages show 9 critical subjects while the Shortage page shows 2 subjects at risk; (d) the subject-wise count of students below 75% differs for some subjects between the Subjects and Shortage pages (for example AOA). These should be confirmed against the underlying measures.

8. Conclusion

The College Attendance Analytics Dashboard brings student, subject and session-level attendance data into an interactive five-page Power BI report. Power Query is used for data preparation, DAX measures calculate overall attendance, student counts, students below 75%, critical students, total classes and the percentage of students at risk, and KPI cards, charts, tables and slicers let users explore the data by academic year, department, subject, gender and date range. The Shortage page groups students into Safe, Warning and Critical categories and estimates the classes required to reach the 75% target, which helps to identify attendance risks and supports follow-up with the students and subjects concerned. The project is a demonstration of attendance monitoring and shortage analysis; further validation of the data model and measures would be advisable before wider use.